

Computer Networks

Why This Matters Now

- CN remains a confirmed weak area from BARC and is directly relevant to HPCL.

High-Yield Topics

- OSI and TCP/IP layers
- Packet, circuit, and virtual-circuit switching
- Error detection and framing basics
- Routing: shortest path, flooding, distance vector, link state
- IP addressing, CIDR, fragmentation, ARP, DHCP, ICMP, NAT
- TCP, UDP, sockets, flow control, congestion control
- Application protocols: DNS, SMTP, HTTP, FTP, email basics

Common Traps

- Subnet mask and host-count mistakes
- Confusing routing protocol behavior
- Wrong fragmentation offsets
- Mixing TCP flow control with congestion control

Done For Now Means

- one subnetting sheet
- one routing protocol comparison sheet
- 15 mixed MCQs with at least 5 numericals